



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4014 Deep Learning II

Final Project

Presentation date: Last 2 class of the last week, Wed and Fri (will be a draw)

Final submission due date: One week after the last class (Friday of week after last week, end of the day)

Project Theme: Object Detection & Generative AI

In this project, students are expected to build or deploy solutions involving object detection or generative AI. The goal is to demonstrate a strong understanding of deep learning techniques and to apply these in a meaningful project.

You may choose ONE of the following project directions:

Option 1: Object Detection (OD)

- Develop a deep learning model capable of detecting at least two different objects in images.

Option 2: Generative AI

Work with and deploy generative AI models such as:

- Stable Diffusion
- GANs (Generative Adversarial Networks)
- Other multimodal or text to image generation models

Option 3: Propose Your Own Idea

Students may work on a self-defined project involving object detection or generative AI. Any custom idea must be discussed and approved by the instructor before beginning.

Application:

Object Detection (OD) is a branch of computer vision. It is a fast-growing field and has a lot of applications. It is hard these days not to notice OD applications around us, smart and self-driving cars, smart phones, security and so many more.

Dataset Requirements:

- You are free to use existing datasets (e.g., from Kaggle, ImageNet) or collect/build your own.
- Annotation tools (like CVAT, LabelImg, etc.) may be necessary if the dataset is not pre-annotated.
- **Minimum requirement:**
 - At least **200 images** per object class (e.g., 200 images with cars, 200 with bicycles)
 - The two objects **do not** need to appear in every image together
- A larger and well-annotated dataset is recommended for better model performance.

Model:

Important: You can't use the pretrained model with an existing object (e.g. you can't present a YOLO or TensorFlow API that is pretrained, HOWEVER you can use those models and perform transfer learning on a different object).

Deployment:

At the end students are expected to build an application or simply deploy the model on a server (this is optional and will have bonus, you won't be marked on this requirement).

Agile development:

You are expected to manage and monitor your development using an agile methodology, i.e. scrum, you can use Jira to do this.

This is required and you need to submit this in your report: Sprints, scrum board, burndown charts etc....

General Requirements**Group Formation:**

- Groups should be finalized by end of Week 1
- Minimum group size: 4
- Maximum group size: 8 (flexible)
- Each group must appoint a Project Manager, who:
 - Coordinates project meetings and timelines
 - Keeps minutes and ensures deliverables are on track
 - Is clearly listed in the report and presentation

Workload Distribution:

- All group members must contribute meaningfully
- Redundancy in tasks (e.g., multiple members training different models) is acceptable but must be clearly documented
- A task distribution table must be included at the start of the final report

Project Report Guidelines

Your report must include the following sections (minimum):

1. Background and Problem Statement
2. Plan of Attack
 - What approach did you take and why?
3. The Dataset
 - Source, size, structure, annotations, EDA (imbalances, outliers, etc.)
4. Model Description
 - Architecture, choice of algorithms, tools used
5. Training & Evaluation
 - Performance (e.g., mAP, precision/recall, F1-score)
 - Validation/test results
 - Confusion matrix, visual results
6. Hyperparameter Tuning
 - What changes were made and their effects?
 - Architecture exploration (if any)
7. Benchmarking
 - Did you compare to a baseline or existing method?
8. Discussion & Reflection
 - What worked, what didn't?
 - What would you do differently?

9. Conclusion
 - Summary of learnings and results
10. Team Contributions Table
 - Clearly outline who did what

Final Submission Includes:

1. Full Report (PDF or DOCX)
2. All Code (organized and runnable)
3. Model Outputs on Sample Inputs (include 5–10 examples with prediction outputs)
4. Presentation Slides
5. Dataset (if custom-made, include annotation files)

Important:

All items must be zipped into a single compressed file and submitted on Blackboard by one team member only. (Please also include your group number in the name of the zip file)

Criterion	Poor	Ok	Good	Amazing	Perfect
Comprehensive Project Design and Implementation	- The project lacks a clear problem statement or objective. - The chosen model type is not appropriate for the problem. - There is no evidence of group roles or responsibilities being defined.	- The project includes a basic problem statement and objective. - A model type is chosen but not well-justified. - Group roles are defined but not clearly documented in the report.	- The project includes a clear problem statement and objective. - The chosen model type is appropriate and justified. - Group roles and responsibilities are clearly documented in the report.	- The project includes a detailed problem statement and objective. - The chosen model type is appropriate, justified, and innovative. - Group roles and responsibilities are clearly documented and reflect effective collaboration	- The project includes a detailed problem statement and objective with real-world relevance. - The chosen model type is appropriate, justified, innovative, and effectively implemented. - Group roles and responsibilities are clearly documented, reflect effective collaboration, and demonstrate leadership and

					initiative.
Effective Data Analysis and Model Evaluation	- There is no evidence of exploratory data analysis (EDA). - Model evaluation metrics are missing or incorrect. - No attempt to address data imbalance or outliers.	- Basic EDA is present but lacks depth (e.g., only class balance is considered). - Model evaluation metrics are present but not well-explained. - Minimal attempt to address data imbalance or outliers.	- EDA includes class balance and correlation matrices. - Model evaluation metrics are present and explained. - Some attempt to address data imbalance and outliers.	- EDA includes class balance, correlation matrices, and additional relevant analyses. - Model evaluation metrics are comprehensive and well-explained. - Effective strategies to address data imbalance and outliers are implemented.	- EDA includes class balance, correlation matrices, and insightful analyses that inform model design. - Model evaluation metrics are comprehensive, well-explained, and include advanced techniques. - Innovative strategies to address data imbalance and outliers are implemented and justified.
Quality of Report and Presentation	- The report is missing key sections or lacks coherence. - The presentation is significantly under or over the time limit. - Visualizations are missing or unclear.	- The report includes all required sections but lacks detail. - The presentation is within the time limit but lacks engagement. - Visualizations are present but	- The report includes all required sections with sufficient detail. - The presentation is within the time limit and engages the audience. - Visualizations are clear	- The report includes all required sections with detailed analysis and insights. - The presentation is within the time limit, engaging, and well-structured. - Visualizations	- The report includes all required sections with detailed analysis, insights, and reflections. - The presentation is within the time limit, highly engaging, and professionally delivered.

		not well-integrated into the report.	and integrated into the report.	ns are clear, integrated, and enhance understanding.	- Visualizations are clear, integrated, enhance understanding, and demonstrate creativity.
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